

GNU Radio Based SDR Receiver

GSOC 2026 Application

Applicant: Diego Toledo Luque

1. About me



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open source space exploration



4th Year student of Mathematics and Computer Science at the University of Malaga, Spain.

Founding member of Málaga Space Team.

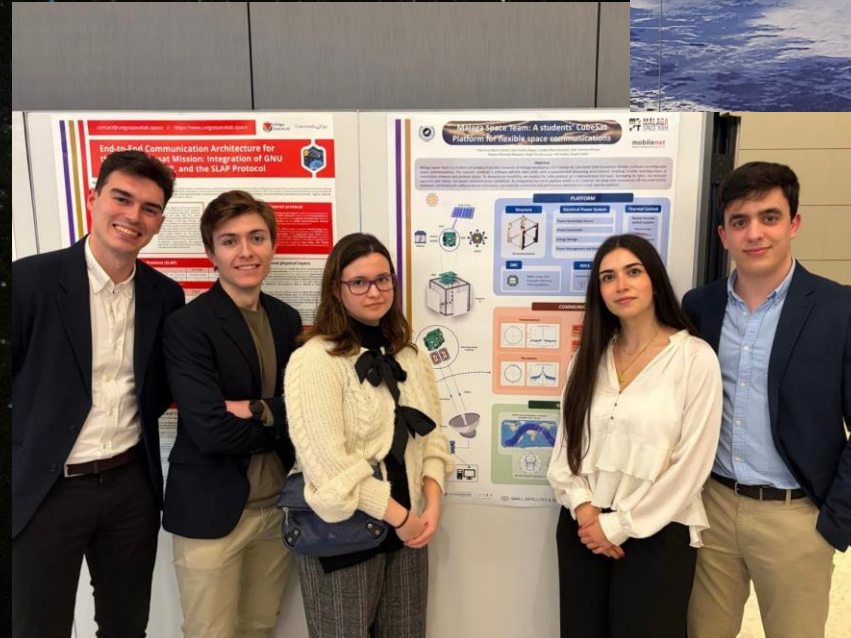
Research Assistant at Mobilenet (Mobile and Aerospace Research Laboratory) from the University of Málaga.

Participated in activities from ESA:

- Cubesat Hands-On Training Week (CHOTW) 2025 at ESEC, Belgium.
- Computer In A Room Challenge 3, online.
- 10th TTC International Workshop at ESOC, Germany.

Participated in Space Challenges 2025 program organized by Endurosat in Sofia, Bulgaria.

Attended Small Satellites and Services International Forum (SSSIF) 2025 and 2026 in Málaga, Spain.



CHOTW 2025
SSSIF 2026

1. About me



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Some space-based projects done:

- AX.25 Encoder – Decoder in GNU Radio
- LoRa modulator – demodulator in GNU Radio ← Poster presented in ESOC
- Fire risk assessment algorithm using Sentinel images
- Currently working with Málaga Space Team building the CubeSAT

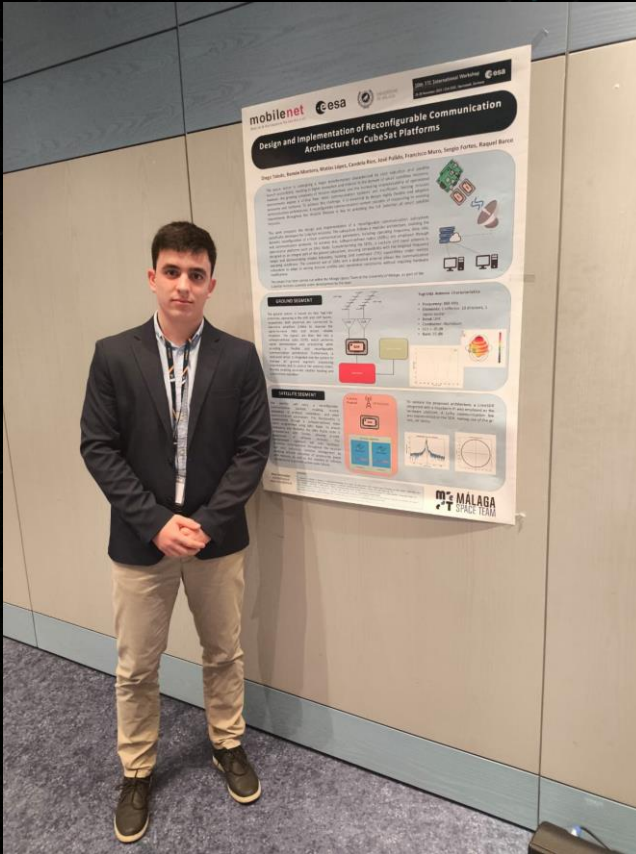


Photo with Jan Wörner in Space Challenges



Photo at ESOC, Germany



2. Málaga Space Team



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MÁLAGA
SPACE TEAM

Created in 2025 as a evolution from K-SAT a different team which had a focus on space entrepenurship.

> 30 students involved

Telecommunications, mechanical, robotics, computer engineers

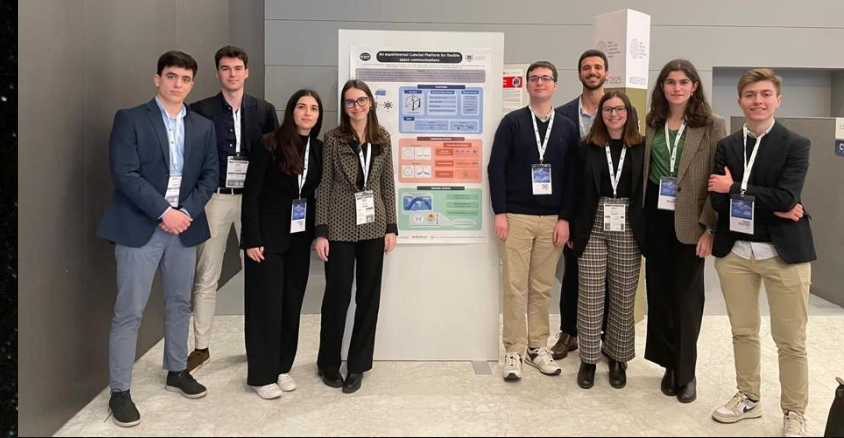
3 hackathon prices

Cassini, NASA Space Apps Challenge

Assistance to several international conferences.

10th TTC International Workshop, SEFUW 2025, SSSIF 2025 & 2026, etc

Assistance to courses organized by European Space Agency.



2. Málaga Space Team

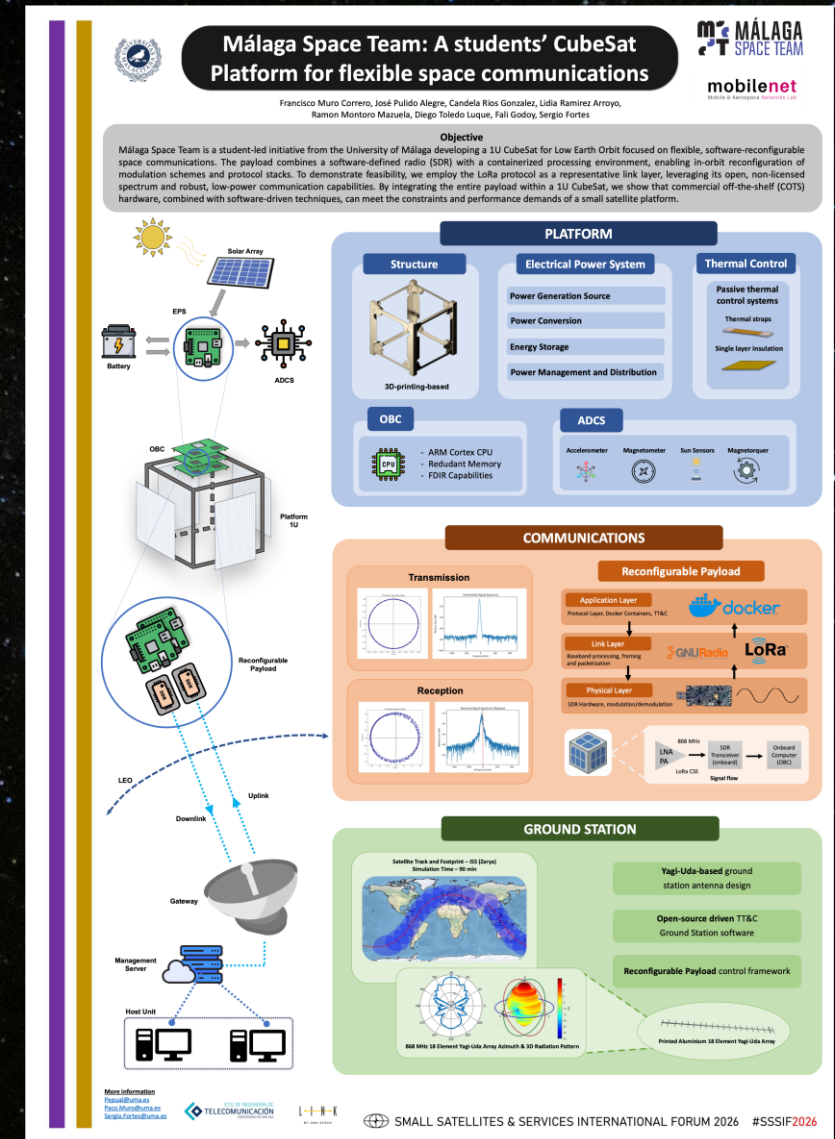


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MÁLAGA
SPACE TEAM

Among the objectives of Málaga Space Team are:

- Promoting space ecosystem among students at the University of Málaga.
- Designing a CubeSat while learning about space along the way.
- Launching the first satellite from the University of Málaga.
- Building a community of highly motivated students in the aerospace sector.



3. Project Proposal



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2 Projects related to my experience:

4. CCSDS GNU Radio Receiver & Decoder

Description: LibreCube is developing an [UHF CubeSat communications board](#). This project will complement this activity by creating a lab-style ground station, utilizing SDR and GNU Radio flowgraphs (and an optional small OOT module) to receive, demodulate, and decode CCSDS packets. Data can come from real SDR and from provided baseband IQ recording. The work will take inspiration from the existing CCSDS GNU Radio decoder (gr-cc_sdr, gr-satellites, gr-ccsds). Deliverables include working flowgraphs, packet/frame decoding, examples and clear documentation with sample datasets.

Skills: C++, GNU Radio, Signal Processing

Proposed Mentors: Tamara (tamara@librecube.org), Shayan (shayan@librecube.org)

Project Size: 350 hours

Difficulty: hard

5. GNU Radio LoRa Receiver with Optional TinyGS / LoRaWAN Integration

Description: This project focuses on creating an SDR-based ground station receiver using GNU Radio capable of receiving and demodulating LoRa satellite signals with standard SDR hardware. The receiver will allow conventional SDR ground stations to decode LoRa packets and optionally forward them to networks such as [TinyGS](#) or [LoRaWAN](#) infrastructure. The work will build on existing GNU Radio LoRa implementations and produce reusable flowgraphs, optional modules, and clear documentation.

Skills: C++, GNU Radio, Signal Processing

Proposed Mentors: Shayan (shayan@librecube.org)

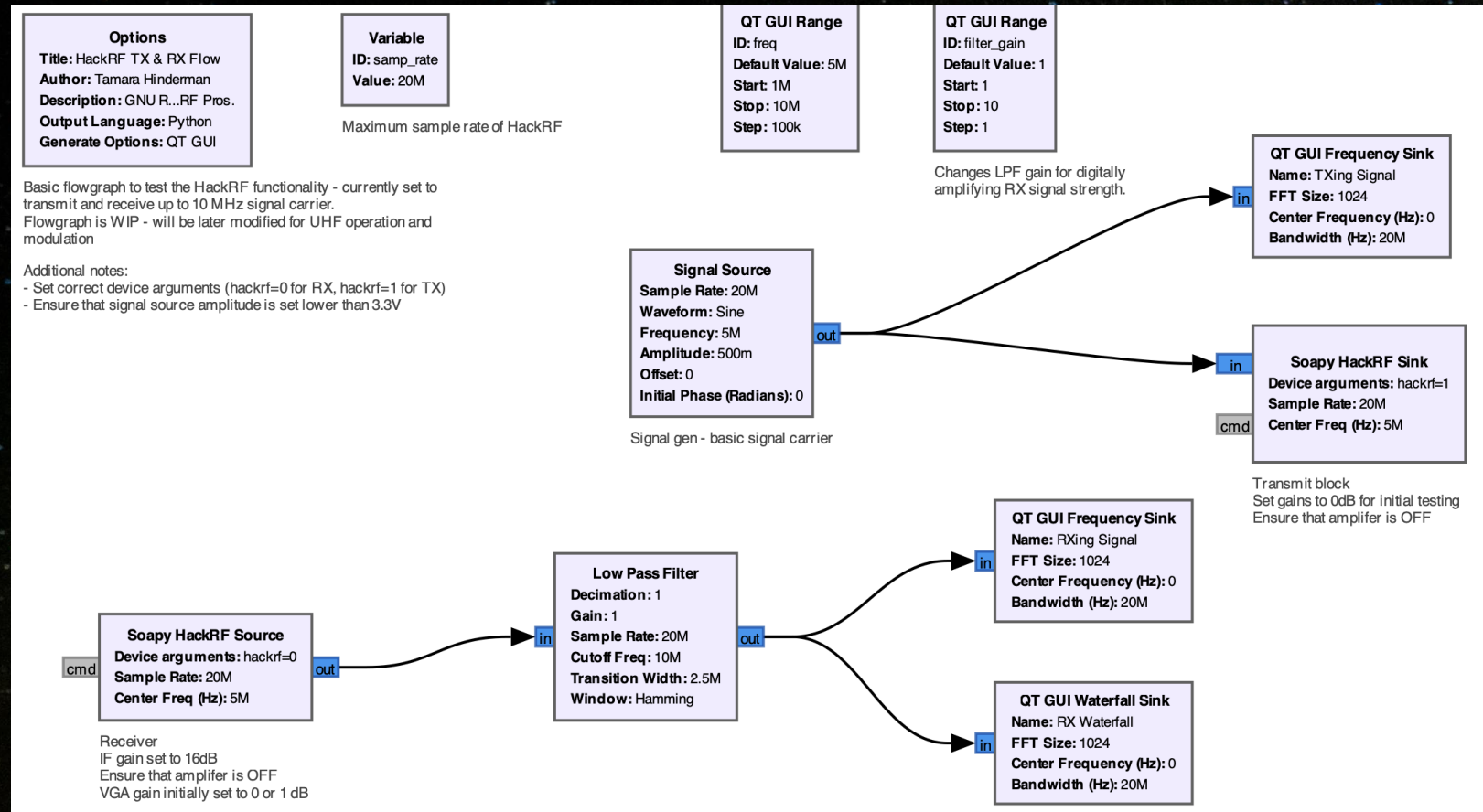
Project Size: 350 hours

Difficulty: hard

3. Project Proposal



Example flowgraph from gitlab repository:



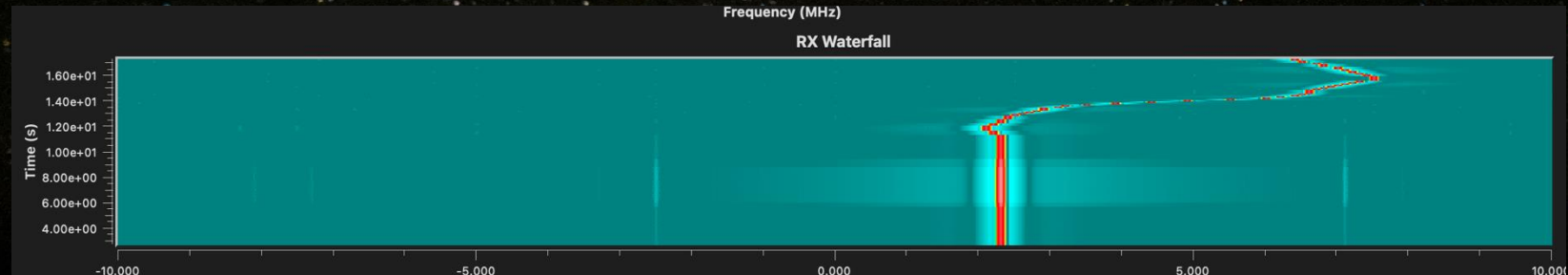
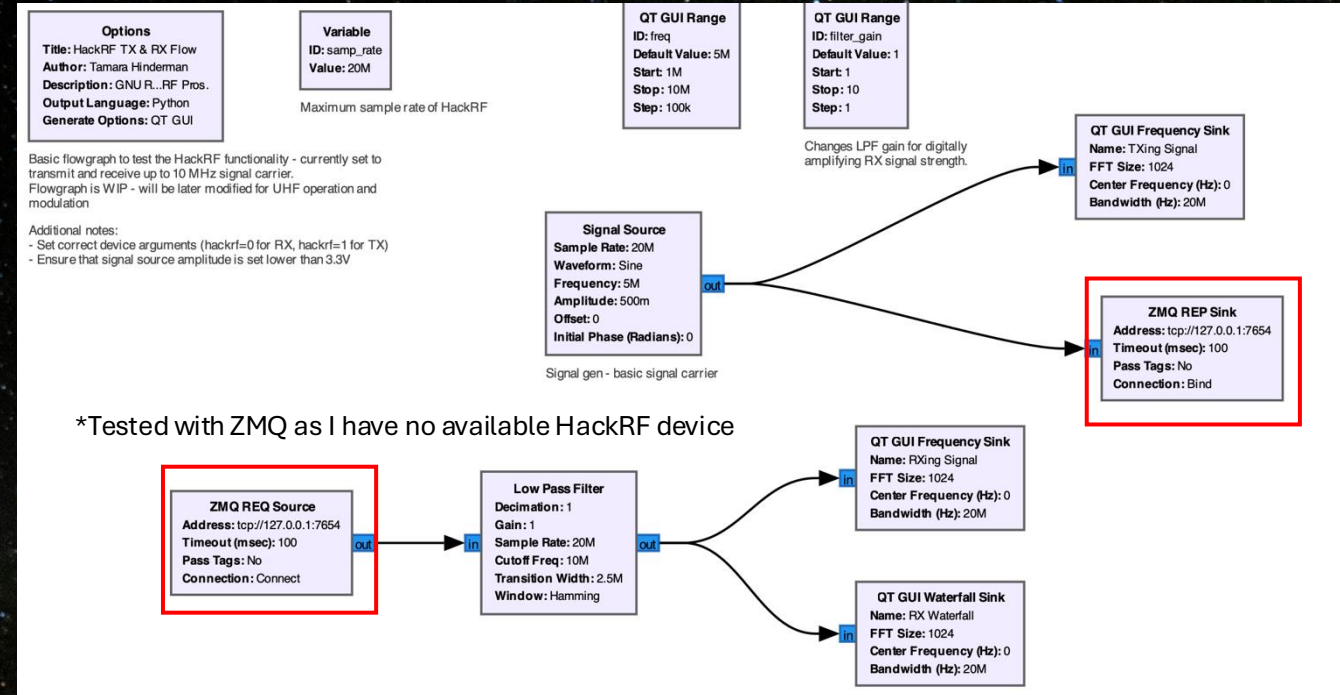
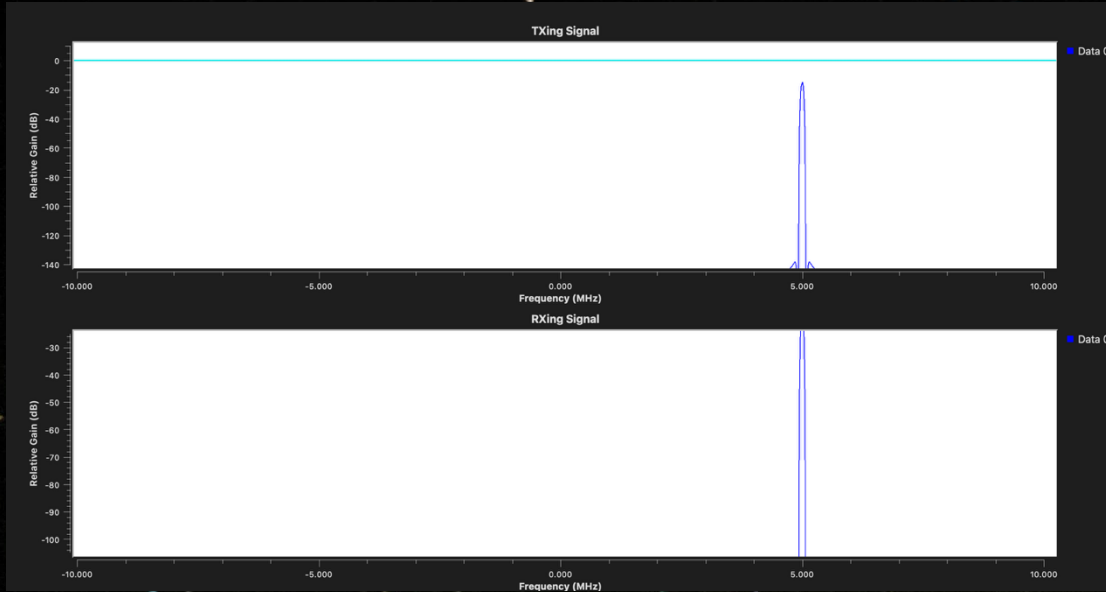
3. Project Proposal



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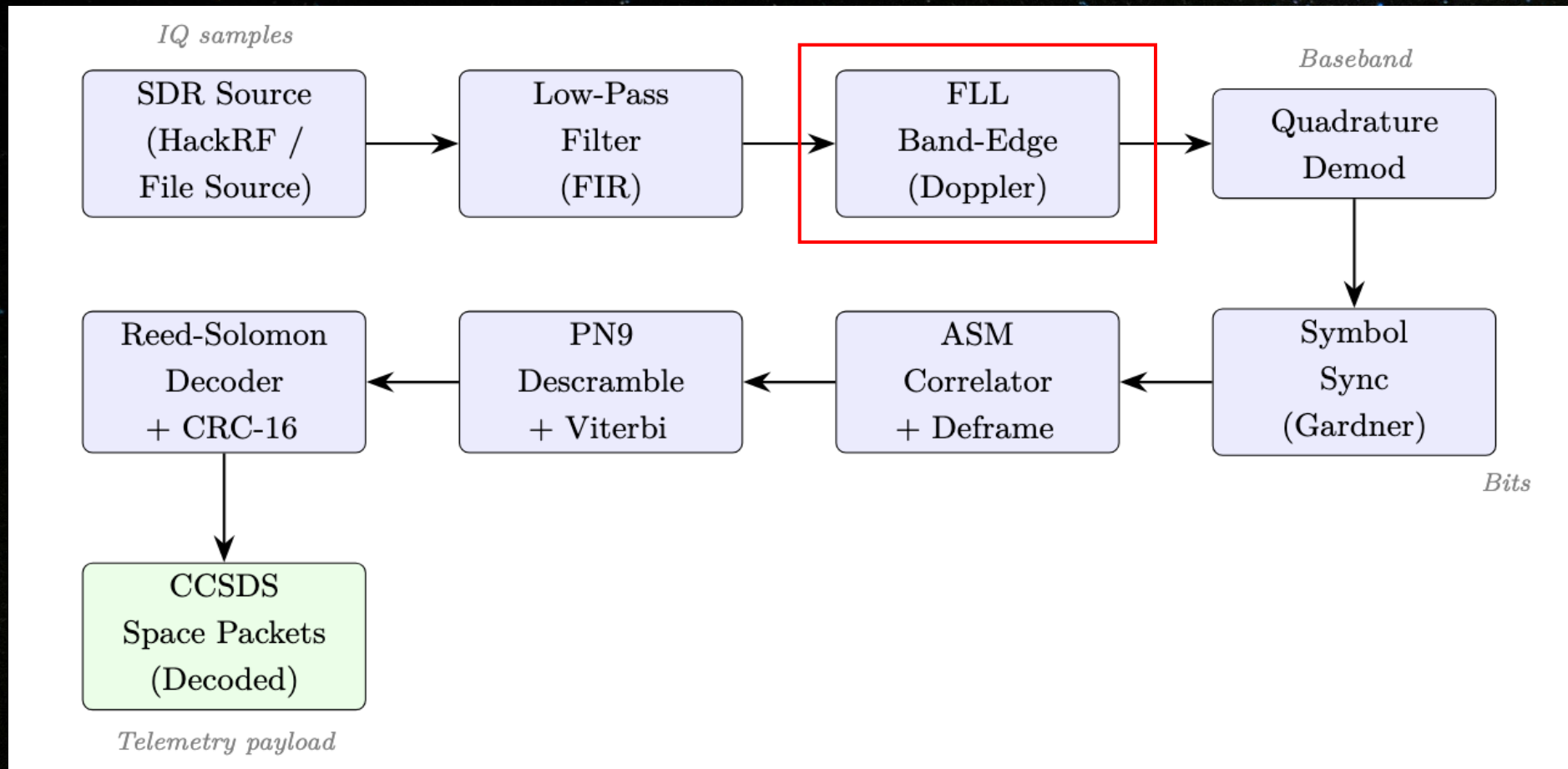
Given flowgraph has been tested:



3. Project Proposal



Final output should follow a similar schema:



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Previous schema has been based on documentation found in the repository:

*modulation?

From CCSDS 130.1-G-3:

“Channel coding is a signal-processing technique by which data can be sent from a source to a destination through a noisy channel so that distinct messages are easily distinguishable from one another.

A channel encoder is then a device that takes this string of binary data and produces a modulating waveform as output.”

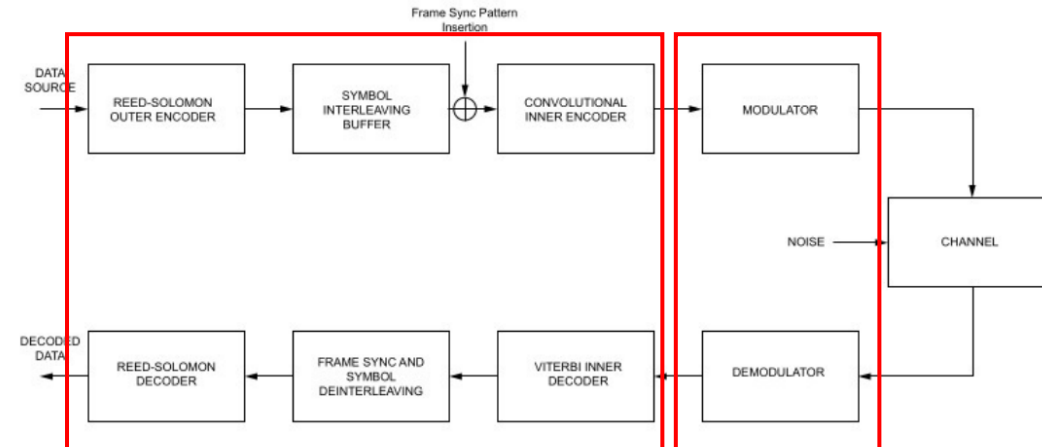


Figure 6-1: Concatenated Coding System Block Diagram

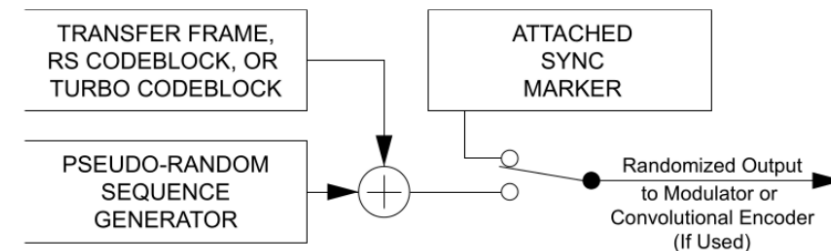


Figure 9-1: Block Diagram of the Recommended Pseudo-Randomizer

3. Project Proposal



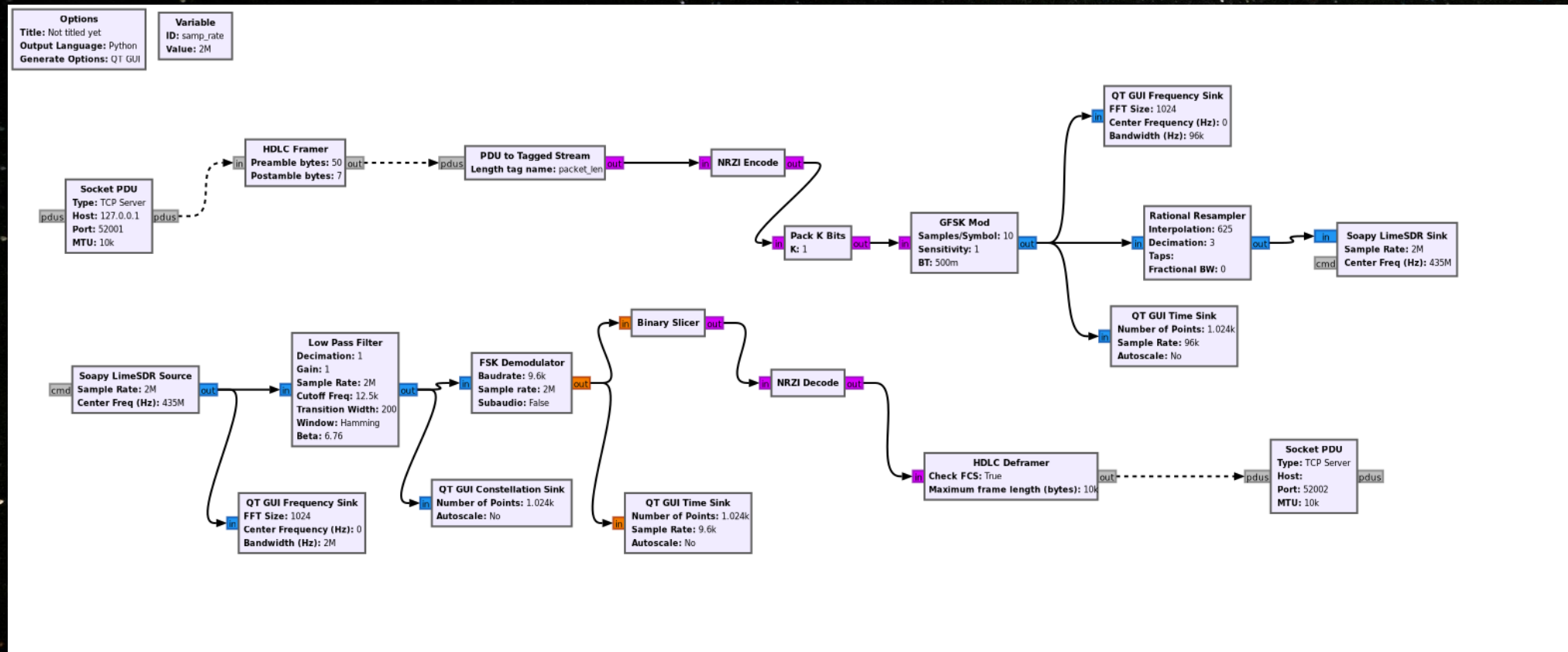
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PREVIOUS WORK:

AX.25 encoder and decoder on GNU Radio

This is part of the communication stack of our CubeSat. Next step is changing modulation to LoRa.



3. Project Proposal



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Phase	Weeks	Hours	Primary Focus & Deliverables
1	1 to 3	80	Research & Environment Setup. Study LC2302 specs with mentors. Set up GNU Radio 3.10+, gr-satellites, gr-ccsds. Write architecture document. Build basic IQ-to-symbol demodulation flowgraph.
2	4 to 6	90	Synchronisation & Deframing. Doppler correction (FLL). Clock recovery (Gardner TED / Symbol Sync). ASM correlator and CCSDS frame deframing. PN9 descrambling.
3	7 to 9	80	FEC Decoding & End-to-End Chain. Viterbi decoder integration. Reed-Solomon (255,223) decoder. CRC-16 validation. Complete IQ-to-payload flowgraph.
4	10 to 11	50	Optional OOT Module & Examples. C++ OOT block for LC2302 telemetry header parsing (if needed). Example scripts and sample flowgraphs. ZMQ output for dashboard integration.
5	12+	50	Documentation & Test Datasets. Curated IQ recordings (clean, AWGN, Doppler). Step-by-step Markdown/Sphinx documentation. Final code cleanup, merge requests, and GSoC report.

4. Questions?



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¿QUESTIONS?





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